

Five top-level non-i.i.d. conditional-independence targets

Choose the target and information set before the nuisance learner or reference distribution.

A Current-state context

$$X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}$$

A1: many subjects, bounded visits [developed]

A2: one/few long sequences [temporal]

A3: both dimensions grow [future]

B Stable subject context

$$X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}, \theta_i$$

Observed baseline or genetics [reduces to A]

Estimated subject context [rate conditions]

Latent sparse context [identification]

C Observed history

$$X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}, H_{it}$$

C1: bounded complete history [reduces to A1]

C2: growing history/filtration [temporal]

Incomplete history changes the null

D Complete trajectories

$$\bar{X}_{it} \perp\!\!\!\perp \bar{Y}_{it} \mid \bar{Z}_{it}$$

Path-level object; scalar visit scores are not an omnibus trajectory test. [future]

E Trajectories + subject context

$$\bar{X}_{it} \perp\!\!\!\perp \bar{Y}_{it} \mid \bar{Z}_{it}, \theta_i$$

Combines path-space calibration with observed or identified subject context. [future]